Wazuh SOC Home Lab Implementation Report

Hands-On SOC Monitoring Lab

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SIEM Home Lab Project Report

Overview:

Built a small-scale SOC lab using Linux Mint as the Wazuh Manager, Kali Linux as an attacker, and Windows 10 as a monitored endpoint. Configured dual network adapters on each VM—Host-Only for internal lab communication and NAT for internet access—and assigned static IPs for reliable connectivity. Set up proper routing so all machines could reach each other and the internet. Deployed Wazuh agents on Kali and Windows, generated test attacks such as SSH login attempts, and monitored them on the Wazuh dashboard. Documented the workflow from event generation to alert triage, including rule IDs, severity, and system behavior, demonstrating practical knowledge of networking, endpoint monitoring, and entry-level SOC operations.

Lab Architecture:

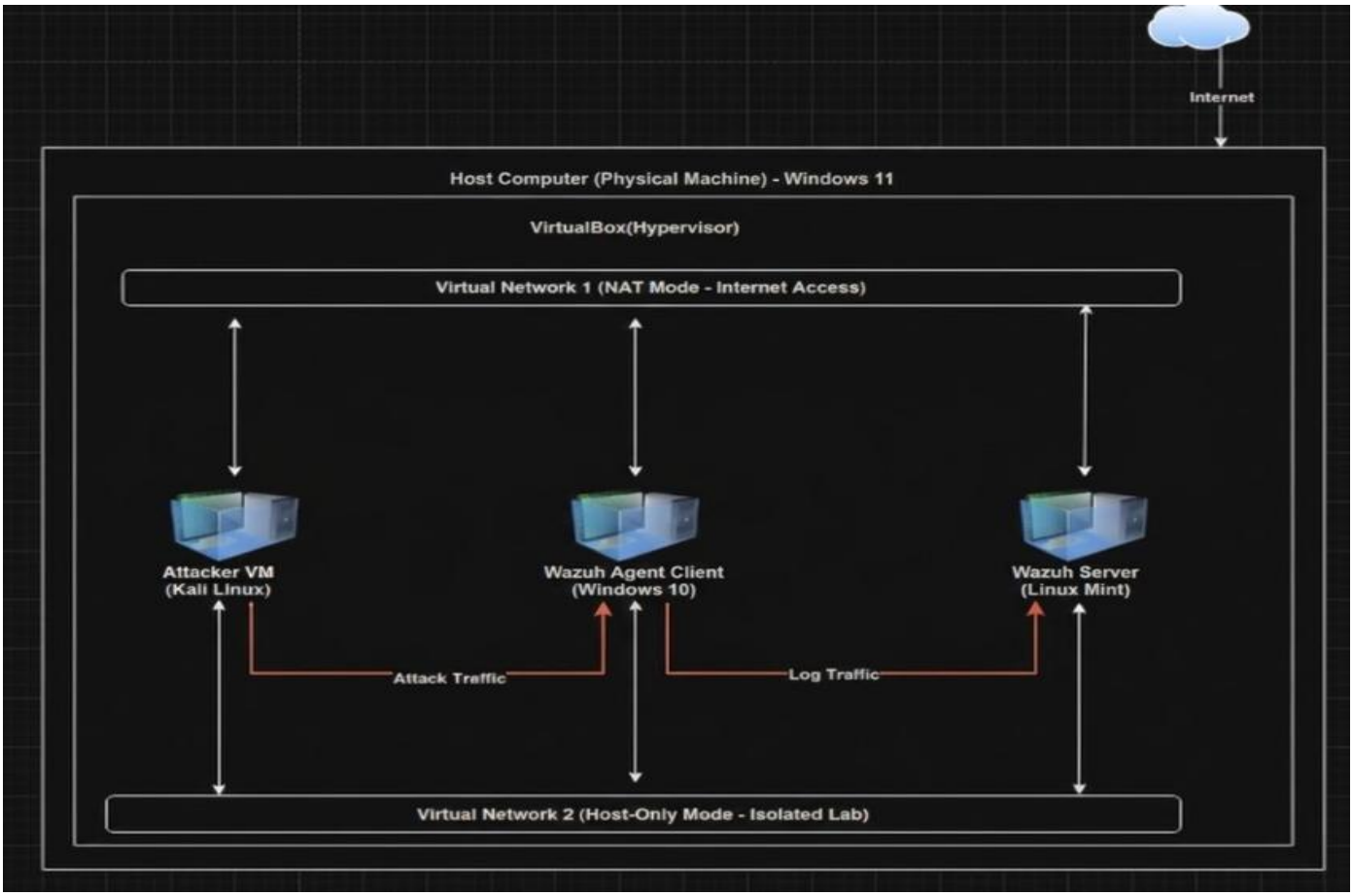
The lab consists of three virtual machines running inside VirtualBox:

1. Linux Mint – Wazuh Manager, Indexer, and Dashboard
2. Kali Linux – Attacker machine
3. Windows 10 – Monitored endpoint

Each machine uses two network interfaces:

• NAT Network Adapter – Provides internet access

• Host-Only Adapter – Allows isolated communication between lab machines



Machine 1 – Wazuh Server (Linux Mint):

System Role

This machine hosts the Wazuh SIEM infrastructure including:

- **Wazuh Manager:**

```
lines 1-23...skipping...
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-09 22:51:49 GMT; 1min 37s ago
    Process: 996 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 231 (limit: 9422)
     Memory: 937.5M (peak: 959.2M)
        CPU: 52.137s
     CGroup: /system.slice/wazuh-manager.service
             ├─1586 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1587 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1588 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1591 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1594 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1642 /var/ossec/bin/wazuh-authd
             ├─1762 /var/ossec/bin/wazuh-db
             ├─2073 /var/ossec/bin/wazuh-execd
             ├─2191 /var/ossec/bin/wazuh-analysisd
             ├─2278 /var/ossec/bin/wazuh-syscheckd
             ├─2306 /var/ossec/bin/wazuh-remoted
             ├─2464 /var/ossec/bin/wazuh-logcollector
             ├─2492 /var/ossec/bin/wazuh-monitord
             └─2531 /var/ossec/bin/wazuh-modulesd

Mar 09 22:51:46 wazuhserver-VirtualBox env[996]: Started wazuh-logcollector...
Mar 09 22:51:46 wazuhserver-VirtualBox env[996]: wazuh-monitord: Process 2485 not used by Wazuh, removing...
Mar 09 22:51:46 wazuhserver-VirtualBox env[996]: Started wazuh-monitord...
Mar 09 22:51:46 wazuhserver-VirtualBox env[996]: wazuh-modulesd: Process 2585 not used by Wazuh, removing...
Mar 09 22:51:46 wazuhserver-VirtualBox env[2523]: 2026/03/09 22:51:46 wazuh-modulesd:router: INFO: Loaded router mod>
Mar 09 22:51:46 wazuhserver-VirtualBox env[2523]: 2026/03/09 22:51:46 wazuh-modulesd:content_manager: INFO: Loaded c>
Mar 09 22:51:46 wazuhserver-VirtualBox env[2523]: 2026/03/09 22:51:46 wazuh-modulesd:inventory-harvester: INFO: Load>
Mar 09 22:51:47 wazuhserver-VirtualBox env[996]: Started wazuh-modulesd...
Mar 09 22:51:49 wazuhserver-VirtualBox env[996]: Completed.
Mar 09 22:51:49 wazuhserver-VirtualBox systemd[1]: Started wazuh-manager.service - Wazuh manager.
~
```

- **Wazuh Indexer**

```
wazuhserver@wazuhserver-VirtualBox:~$ sudo systemctl status wazuh-indexer
● wazuh-indexer.service - wazuh-indexer
     Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-09 22:51:58 GMT; 2min 16s ago
       Docs: https://documentation.wazuh.com
   Main PID: 994 (java)
      Tasks: 87 (limit: 9422)
     Memory: 1.5G (peak: 1.5G)
        CPU: 42.917s
     CGroup: /system.slice/wazuh-indexer.service
             └─994 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dope

Mar 09 22:51:38 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: System::setSecurityManager has been called
Mar 09 22:51:38 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: Please consider reporting this to the maint
Mar 09 22:51:38 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: System::setSecurityManager will be removed
Mar 09 22:51:39 wazuhserver-VirtualBox systemd-entrypoint[994]: Mar 09, 2026 10:51:39 PM sun.util.locale.provider.Lo
Mar 09 22:51:39 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: COMPAT locale provider will be removed in a
Mar 09 22:51:40 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: A terminally deprecated method in java.lang
Mar 09 22:51:40 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: System::setSecurityManager has been called
Mar 09 22:51:40 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: Please consider reporting this to the maint
Mar 09 22:51:40 wazuhserver-VirtualBox systemd-entrypoint[994]: WARNING: System::setSecurityManager will be removed
Mar 09 22:51:58 wazuhserver-VirtualBox systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
lines 1-21/21 (END)
```

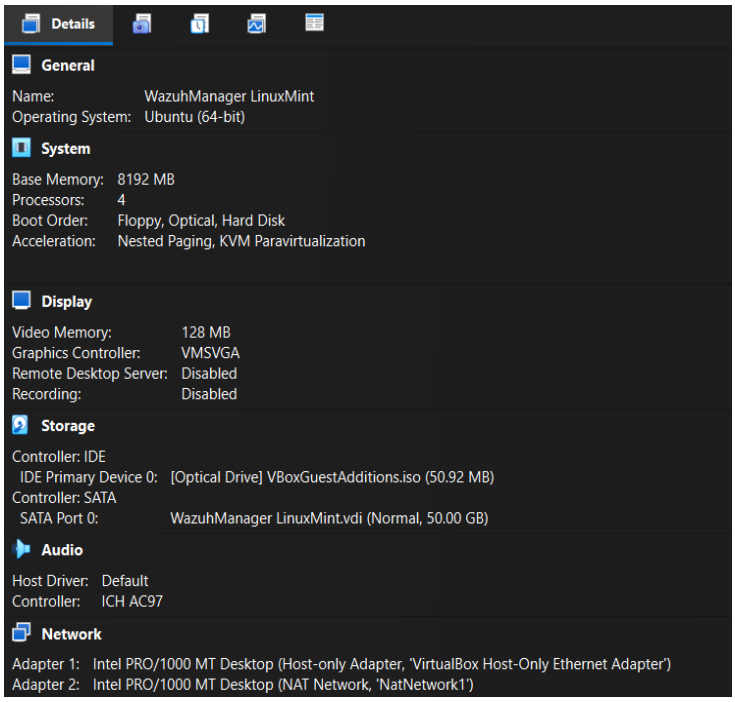
- **Wazuh Dashboard**

```
wazuhserver@wazuhserver-VirtualBox:~$ sudo systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
     Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-09 22:51:27 GMT; 3min 40s ago
   Main PID: 672 (node)
      Tasks: 11 (limit: 9422)
     Memory: 299.9M (peak: 391.9M)
        CPU: 13.122s
     CGroup: /system.slice/wazuh-dashboard.service
             └─672 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-r>

Mar 09 22:52:00 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:00Z">
Mar 09 22:52:01 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:01Z">
Mar 09 22:52:01 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:01Z">
Mar 09 22:52:01 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:01Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
Mar 09 22:52:02 wazuhserver-VirtualBox opensearch-dashboards[672]: {"type":"log","@timestamp":"2026-03-09T22:52:02Z">
lines 1-20/20 (END)
```

Virtual Machine Settings

- OS: Linux Mint
- RAM: 8GB
- CPU: 4
- Storage: 50GB



Machine 2 – Kali Linux (Attacker Machine):

System Role

This system simulates attacker behavior to generate security events for the SIEM.

Virtual Machine Settings

- OS: Kali Linux
- RAM: 3GB
- CPU: 3
- Storage: 50GB

Wazuh Agent Installation

Commands used:

- wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agentamd64.deb
  sudo WAZUH_MANAGER='xyz' WAZUH_AGENT_NAME='KaliAttacker' dpkg -i wazuh-agent.deb

Enable service:

- sudo systemctl enable wazuh-agent
- sudo systemctl start wazuh-agent

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status wazuh-agent
[sudo] password for kali:
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; p>
     Active: active (running) since Mon 2026-03-09 18:57:05 EDT; 1min 42s ago
 Invocation: 886fb3ed8dc5411ea30fad371089b98c
    Process: 840 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (>
      Tasks: 29 (limit: 3351)
     Memory: 113.9M (peak: 119.3M)
        CPU: 9.323s
     CGroup: /system.slice/wazuh-agent.service
             ├─ 894 /var/ossec/bin/wazuh-execd
             ├─ 914 /var/ossec/bin/wazuh-agentd
             ├─ 995 /var/ossec/bin/wazuh-syscheckd
             ├─1065 /var/ossec/bin/wazuh-logcollector
             └─1074 /var/ossec/bin/wazuh-modulesd

Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-s>
Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-a>
Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-e>
Mar 09 18:57:01 kali env[840]: Started wazuh-execd...
Mar 09 18:57:02 kali env[840]: Started wazuh-agentd...
Mar 09 18:57:03 kali env[840]: Started wazuh-syscheckd...
Mar 09 18:57:03 kali env[840]: Started wazuh-logcollector...
Mar 09 18:57:03 kali env[840]: Started wazuh-modulesd...
Mar 09 18:57:05 kali env[840]: Completed.
Mar 09 18:57:05 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.
lines 1-25/25 (END)...skipping...
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-09 18:57:05 EDT; 1min 42s ago
 Invocation: 886fb3ed8dc5411ea30fad371089b98c
    Process: 840 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 29 (limit: 3351)
     Memory: 113.9M (peak: 119.3M)
        CPU: 9.323s
     CGroup: /system.slice/wazuh-agent.service
             ├─ 894 /var/ossec/bin/wazuh-execd
             ├─ 914 /var/ossec/bin/wazuh-agentd
             ├─ 995 /var/ossec/bin/wazuh-syscheckd
             ├─1065 /var/ossec/bin/wazuh-logcollector
             └─1074 /var/ossec/bin/wazuh-modulesd

Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-syscheckd-950.pid' not used...
Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-agentd-901.pid' not used...
Mar 09 18:56:59 kali env[840]: Deleting PID file '/var/ossec/var/run/wazuh-execd-893.pid' not used...
Mar 09 18:57:01 kali env[840]: Started wazuh-execd...
Mar 09 18:57:02 kali env[840]: Started wazuh-agentd...
```

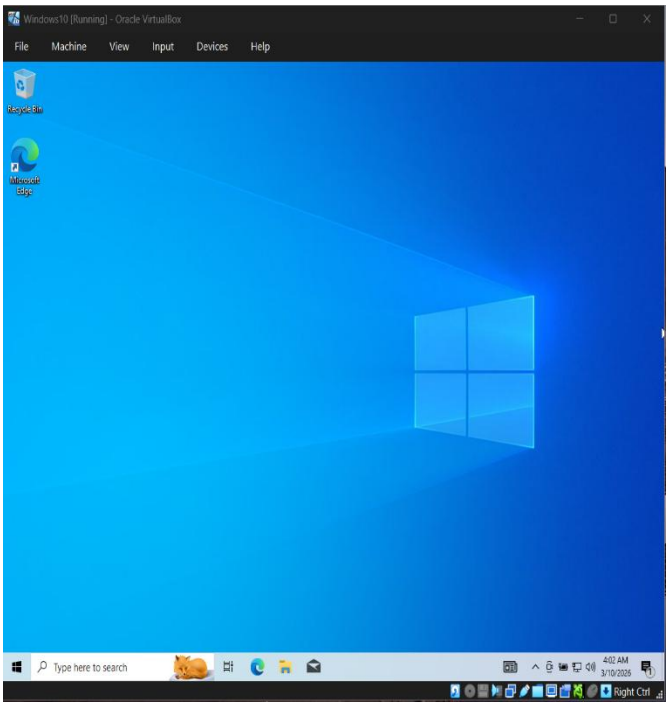
Machine 3 – Windows 10 (Monitored Endpoint):

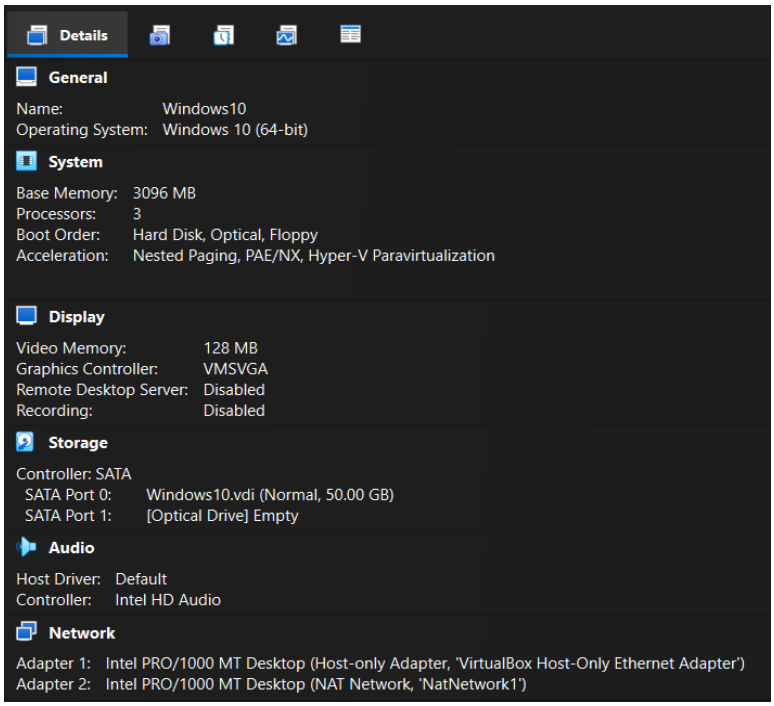
System Role

This system acts as a monitored workstation generating logs for analysis.

Virtual Machine Settings

- OS: Linux Mint
- RAM: 3GB
- CPU: 3
- Storage: 50GB





Wazuh Deployment:

The Wazuh platform consists of three main components that work together to monitor, analyze, and visualize security events across systems. These components include the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard.

1) Wazuh Manager

The Wazuh Manager acts as the central component of the platform. It is responsible for collecting logs from different systems and agents connected to the network. After collecting the logs, the manager performs rule-based analysis to detect suspicious or malicious activities. Based on this analysis, it generates alerts that help security analysts identify potential threats and incidents.

2) Wazuh Indexer

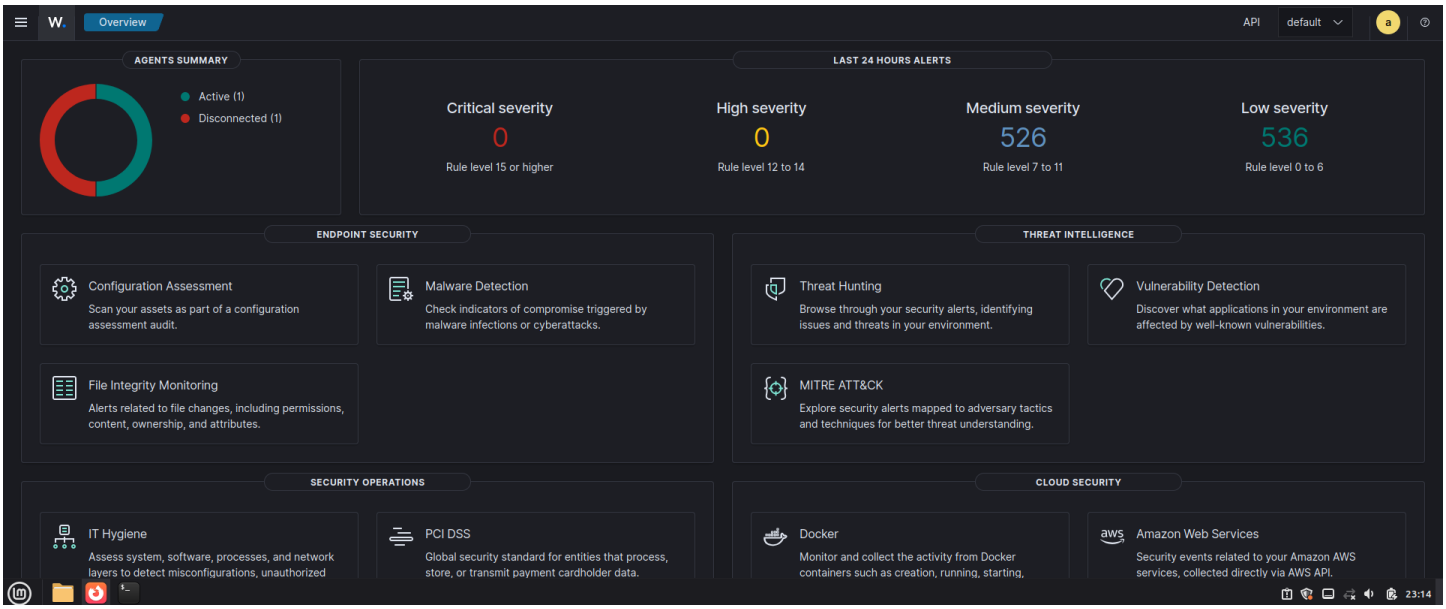
The Wazuh Indexer is responsible for storing and organizing the event data generated by the Wazuh Manager. It indexes the collected logs and alerts so they can be efficiently searched and analyzed. This allows security teams to quickly investigate events, perform threat hunting, and review historical security data.

3) Wazuh Dashboard

The Wazuh Dashboard provides a graphical user interface that allows users to interact with the Wazuh platform. Through the dashboard, users can visualize alerts, monitor logs, review security events, and analyze system activity in an organized and user-friendly way.

Dashboard Access:

The Wazuh Dashboard can be accessed through a web browser using HTTPS. It operates on port 443, which ensures secure communication between the user and the dashboard interface.



Attack Simulation – SSH Brute Force:

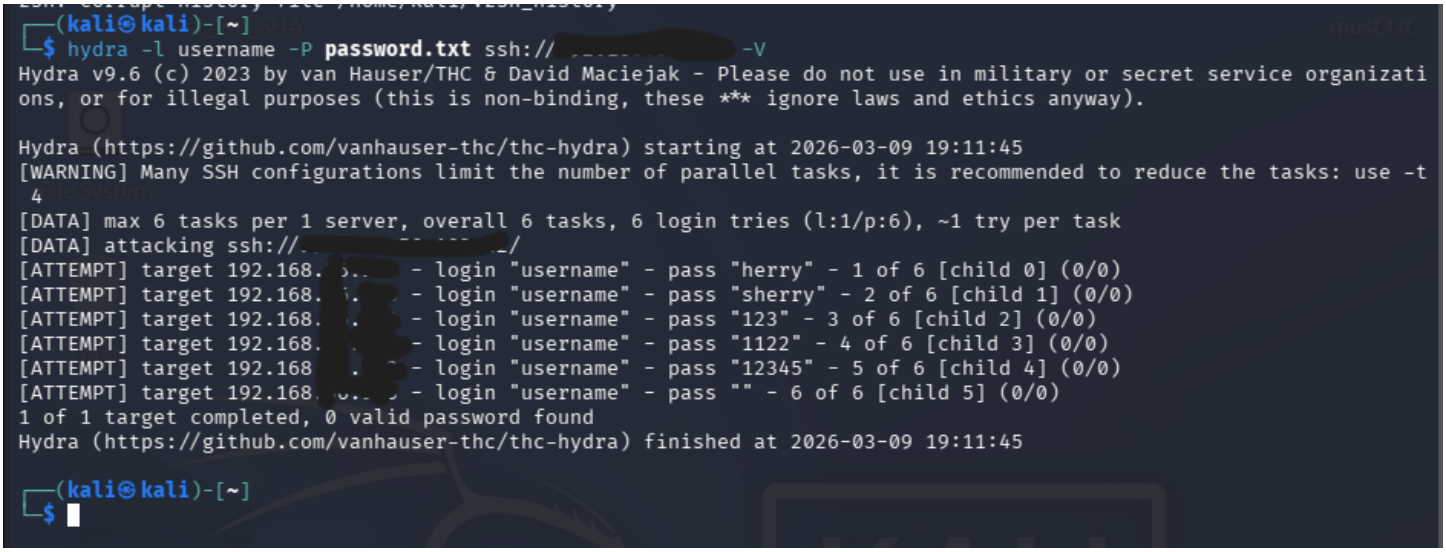
To generate security alerts, an SSH brute force attack was simulated from Kali Linux to the Windows 10 Virtual Machine.

Attack Tool

Hydra (SSH brute force tool)
Example command:
hydra -l username -P passwords.txt ssh://**Target IP**
This command attempts multiple password combinations against the SSH service.

Attack Process

1. Kali sends repeated login attempts to the Windows system.
2. Windows logs authentication events.
3. The Wazuh agent collects these logs.
4. Logs are sent to the Wazuh Manager.
5. Wazuh rules detect suspicious login activity.
6. Alerts are generated and displayed in the dashboard.



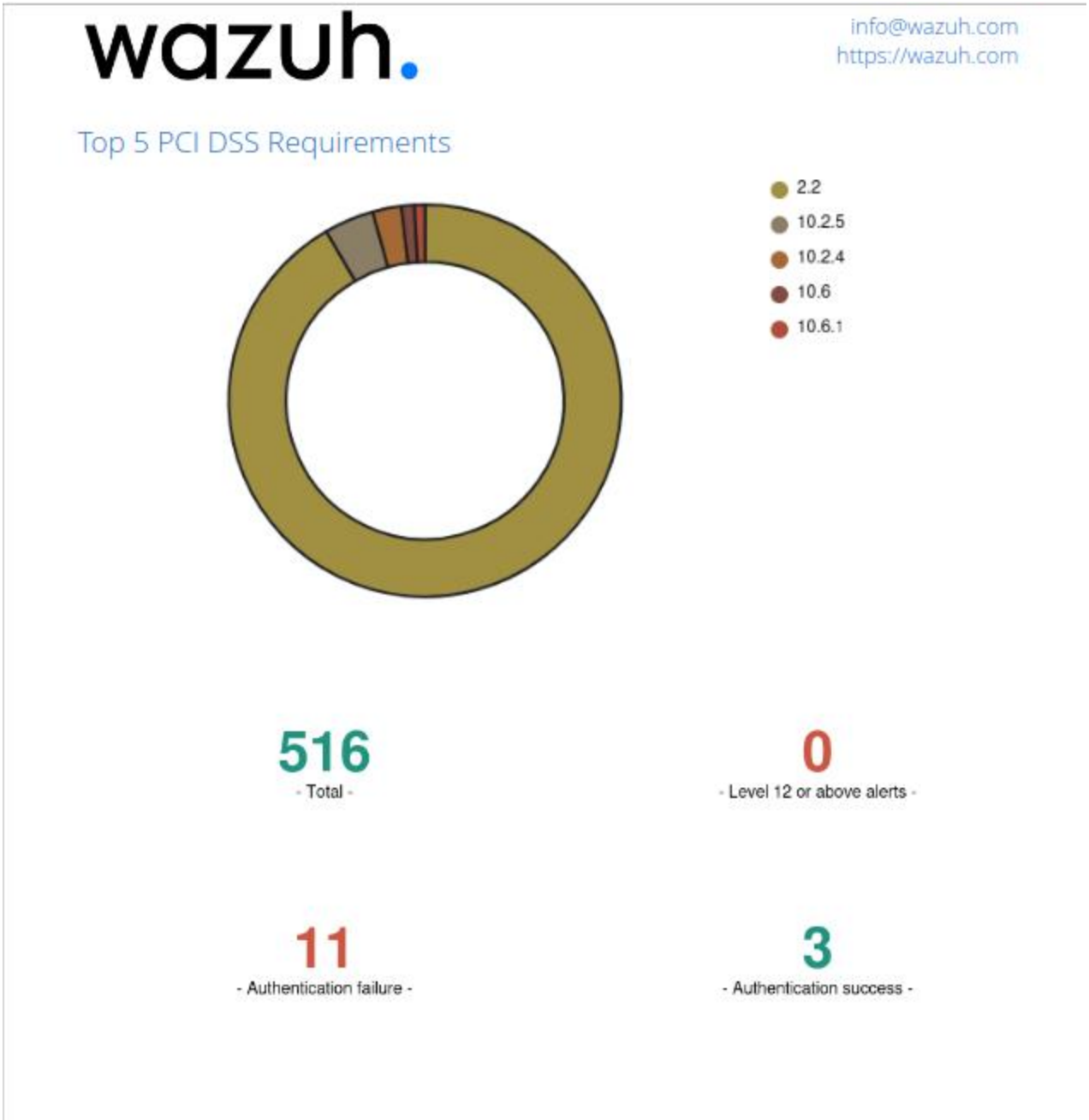
```
┌──(kali㉿kali)-[~]
└─$ hydra -l username -P password.txt ssh:// [redacted] -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
ons, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-09 19:11:45
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t
 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1 try per task
[DATA] attacking ssh:// [redacted] /
[ATTEMPT] target 192.168. [redacted] - login "username" - pass "herry" - 1 of 6 [child 0] (0/0)
[ATTEMPT] target 192.168. [redacted] - login "username" - pass "sherry" - 2 of 6 [child 1] (0/0)
[ATTEMPT] target 192.168. [redacted] - login "username" - pass "123" - 3 of 6 [child 2] (0/0)
[ATTEMPT] target 192.168. [redacted] - login "username" - pass "1122" - 4 of 6 [child 3] (0/0)
[ATTEMPT] target 192.168 [redacted] - login "username" - pass "12345" - 5 of 6 [child 4] (0/0)
[ATTEMPT] target 192.168 [redacted] - login "username" - pass "" - 6 of 6 [child 5] (0/0)
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-09 19:11:45

┌──(kali㉿kali)-[~]
└─$
```

Alert Analysis in Wazuh:

After the attack simulation, the following information was visible in the dashboard:



wazuh.

info@wazuh.com
https://wazuh.com

Groups summary

Groups	Count
sca	426
windows	86
WEF	47
windows_security	19
windows_application	14
authentication_failed	11
windows_system	6
policy_changed	5
ossec	4
account_changed	3
authentication_success	3
adduser	2
group_changed	1
win_group_changed	1

Conclusion

This project demonstrates the deployment of a functional SIEM home lab capable of collecting logs, detecting suspicious activity, and visualizing security alerts. The lab simulates a basic SOC environment with attacker activity, monitored endpoints, and centralized analysis using Wazuh. The project highlights practical skills in virtualization, networking, SIEM deployment, agent configuration, and security event analysis.